
Calculators and communication devices are not allowed.

Answer the following questions.

1. (a) (10 pts.) Let X be a nonzero vector in $\mathbb{R}^n$. Show that $U = \frac{1}{\|X\|}X$ is a unit vector in the direction of X .
(b) Let $S = \{X, Y, Z\}$ be a subset of $\mathbb{R}^4$, where $X = (2, -1, 1, 3)$, $Y = (-2, 1, -1, -3)$ and $Z = (-4, 2, 1, -6)$.
(i) (5 pts.) Find the cosine of the angle θ between X and Y .
(ii) (5 pts.) Find the largest possible value for $\|\sqrt{15}X - \sqrt{15}Y + \sqrt{57}Z\|$.
(iii) (5 pts.) Is $U = (2, -1, 1, 3)$ in $\text{span}(S)$? Justify your answer.
2. (a) (10 pts.) Let X, Y and Z be vectors in $\mathbb{R}^n$. Show that $X \cdot (Y + Z) = X \cdot Y + X \cdot Z$.
(b) (10 pts.) Let A be an $m \times n$ matrix. Show that the null space of A is a subspace of $\mathbb{R}^n$.
(c) Let $A = \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.
(i) (4 pts.) Find $\text{rank}(A)$, $\text{nullity}(A)$, $\text{rank}(A^T)$, $\text{nullity}(A^T)$.
(ii) (2 pts.) Find a basis for the null space of A .
(iii) (2 pts.) Find a basis for the row space of A^T .
(iv) (2 pts.) Find a basis for the row space of A .
3. (a) (10 pts.) Let $S = \{X_1, X_2, \dots, X_n\}$ be a basis for a vector space V . Show that every vector X in V can be expressed, in one and only one way, as a linear combination of the vectors in S .
(b) (10 pts.) Let V be a real vector space. Show that $0X = O$ for any $X \in V$.
(c) (10 pts.) Determine whether the subset $W = \{(a, a + \sqrt{2}, 3a) : a \in \mathbb{R}\}$ of $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$.

4. In each of the following questions, select the correct answer. (3 pts. each)

(a) If X, Y and Z are linearly independent vectors in a vector space V , then

- (A) $2X, 2Y, 2Z$ are linearly independent
- (B) $X, 2y, 3z$ are linearly dependent
- (C) $X, Y, X + Y, Z$ are linearly independent
- (D) None of the above

(b) If $\{(a, b, c), (a', b', c'), (a'', b'', c'')\}$ spans $\mathbb{R}^3$, then

- (A) $\{(a, b, c), (a', b', c'), (a'', b'', c'')\}$ is a basis for $\mathbb{R}^3$
- (B) $\{(a, b, 0, c), (a', b', 0, c'), (a'', b'', 0, c''), (2, 2, 2, 2)\}$ spans $\mathbb{R}^4$
- (C) All of the above
- (D) None of the above

(c) Let W be a subspace of a finite-dimensional vector space V . Then

- (A) $\dim(W) \leq \dim(V)$
- (B) Any basis for W is contained in only one basis for V
- (C) Any linearly independent set of vectors in V is a basis for W .
- (D) None of the above

(d) Let X and Y be vectors in $\mathbb{R}^n$, then

- (A) $\|X - Y\| \leq \|X\| - \|Y\|$
- (B) $\|X - Y\| \leq \|X\| + \|Y\|$
- (C) $\|X - Y\| = \|X\| + \|Y\|$
- (D) None of the above

(e) There exist vectors X and Y in $\mathbb{R}^n$ such that $\|X\| = \frac{3}{2}$ and $\|Y\| = 3$ and

- (A) $X \cdot Y = 5$
- (B) $X \cdot Y = -5$
- (C) $|X \cdot Y| = \|X\| + \|Y\|$
- (D) None of the above

$$1. \quad (a) \quad \|U\| = \left\| \frac{1}{\|X\|} X \right\| = \frac{1}{\|X\|} \|X\| = 1 \implies U \text{ is a unit vector. } \cos \theta = \frac{U \cdot X}{\|U\| \|X\|} = \frac{\left(\frac{1}{\|X\|} X \right) \cdot X}{\|U\| \|X\|} = 1 \implies \theta = 0.$$

$$(i) \quad \|X\| = \sqrt{15}, \|Y\| = \sqrt{15}, X \cdot Y = -15. \implies \cos \theta = -1.$$

$$(ii) \quad \|\sqrt{15}X - \sqrt{15}Y + \sqrt{57}Z\| \leq \|\sqrt{15}X\| + \|\sqrt{15}Y\| + \|\sqrt{57}Z\| = \sqrt{15}\|X\| + \sqrt{15}\|Y\| + \sqrt{57}\|Z\| = 15 + 15 + 57 = 87 \implies 87 \text{ is the largest possible value of } \|\sqrt{15}X - \sqrt{15}Y + \sqrt{57}Z\|.$$

$$(iii) \quad c_1X + c_2Y + c_3Z = (2, -1, 1, 3) \implies \left[\begin{array}{ccc|c} 2 & -2 & -4 & 2 \\ -1 & 1 & 2 & -1 \\ 1 & -1 & 1 & 1 \\ 3 & -3 & -6 & 3 \end{array} \right] \equiv \dots \equiv \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \quad c_1 = r + 1, c_2 = r, c_3 = 0 \text{ (Or } U = X, \text{ Hence, yes).}$$

Yes.

$$2. \quad (a) \quad \text{Let } X = (x_1, x_2, \dots, x_n), Y = (y_1, y_2, \dots, y_n), Z = (z_1, z_2, \dots, z_n) \text{ be vectors in } \mathbb{R}^n. \text{ Then } X \cdot (Y + Z) = X \cdot (y_1 + z_1, y_2 + z_2, \dots, y_n + z_n) = x_1(y_1 + z_1) + x_2(y_2 + z_2) + \dots + x_n(y_n + z_n) = (x_1y_1 + x_2y_2 + \dots + x_ny_n) + (x_1z_1 + x_2z_2 + \dots + x_nz_n) = X \cdot Y + X \cdot Z.$$

$$(b) \quad \text{Let } A \text{ be an } m \times n \text{ matrix. Let } W = \{X \in \mathbb{R}^n : AX = O_{m \times 1}\}.$$

$$(1) \quad AO_{n \times 1} = O_{m \times 1} \implies O_{n \times 1} \in W \implies W \neq \emptyset.$$

$$(\alpha) \quad \text{Let } X, Y \in W \implies AX = O, AY = O. A(X + Y) = AX + AY = O + O = O \implies X + Y \in W.$$

$$(\beta) \quad \text{Let } k \in \mathbb{R}, X \in W \text{ thus, } AX = O. A(kX) = k(AX) = kO = O \implies kX \in W.$$

$$(c) \quad A = \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \equiv \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(i) \quad \text{rank}(A) = 2, \text{nullity}(A) = 2, \text{rank}(A^T) = 2, \text{nullity}(A^T) = 1.$$

$$(ii) \quad \text{Basis for the null space of } A = \{(1, 0, 0, 0), (0, 0, 1, 0)\}$$

$$(iii) \quad \text{Basis for the row space of } A^T = \{(0, 1, 0), (-1, 0, 1)\}$$

$$(iv) \quad \text{Basis for the row space of } A = \{(0, 1, 0, 0), (0, 0, 0, 1)\}$$

$$3. \quad (a) \quad \text{Let } S = \{X_1, X_2, \dots, X_n\} \text{ be a basis for a vector space } V. \text{ Suppose } X \in V \text{ such that } X = c_1X_1 + c_2X_2 + \dots + c_nX_n \text{ for some } c_1, c_2, \dots, c_n \in \mathbb{R} \text{ and } X = d_1X_1 + d_2X_2 + \dots + d_nX_n \text{ for some } d_1, d_2, \dots, d_n \in \mathbb{R} \implies O = X - X = (c_1 - d_1)X_1 + (c_2 - d_2)X_2 + \dots + (c_n - d_n)X_n. \text{ Since } S \text{ is linearly independent, then } (c_1 - d_1) = (c_2 - d_2) = \dots = (c_n - d_n) = 0 \implies c_1 = d_1, c_2 = d_2, \dots, c_n = d_n.$$

$$(b) \quad 0X = (0 + 0)X = 0X + 0X. \text{ Since } 0X \in V, \text{ then } -0X \in V. \underline{0} = 0X + (-0X) = (0X + 0X) + (-0X) = 0X + [0X + (0X)] = 0X + O = \underline{0X}$$

$$(c) \quad W \text{ is not a subspace of } \mathbb{R}^3, \text{ for } O \notin W, \text{ or } (1, 1 + \sqrt{2}, 3) + (2, 2 + \sqrt{2}, 6) = (3, 3 + 2\sqrt{2}, 9) \notin W.$$

4. In each of the following questions, select the correct answer. (3 pts. each)

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(A) $2X, 2Y, 2Z$ are linearly independent
- (b) If $\{(a, b, c), (a', b', c'), (a'', b'', c'')\}$ spans $\mathbb{R}^3$, then
(C) All of the above
- (c) Let W be a subspace of a finite-dimensional vector space V . Then
(A) $\dim(W) \leq \dim(V)$
- (d) Let X and Y be vectors in $\mathbb{R}^n$, then
(B) $\|X - Y\| \leq \|X\| + \|Y\|$
- (e) There exist vectors X and Y in $\mathbb{R}^n$ such that $\|X\| = \frac{3}{2}$ and $\|Y\| = 3$ and
(C) $|X \cdot Y| = \|X\| + \|Y\|$